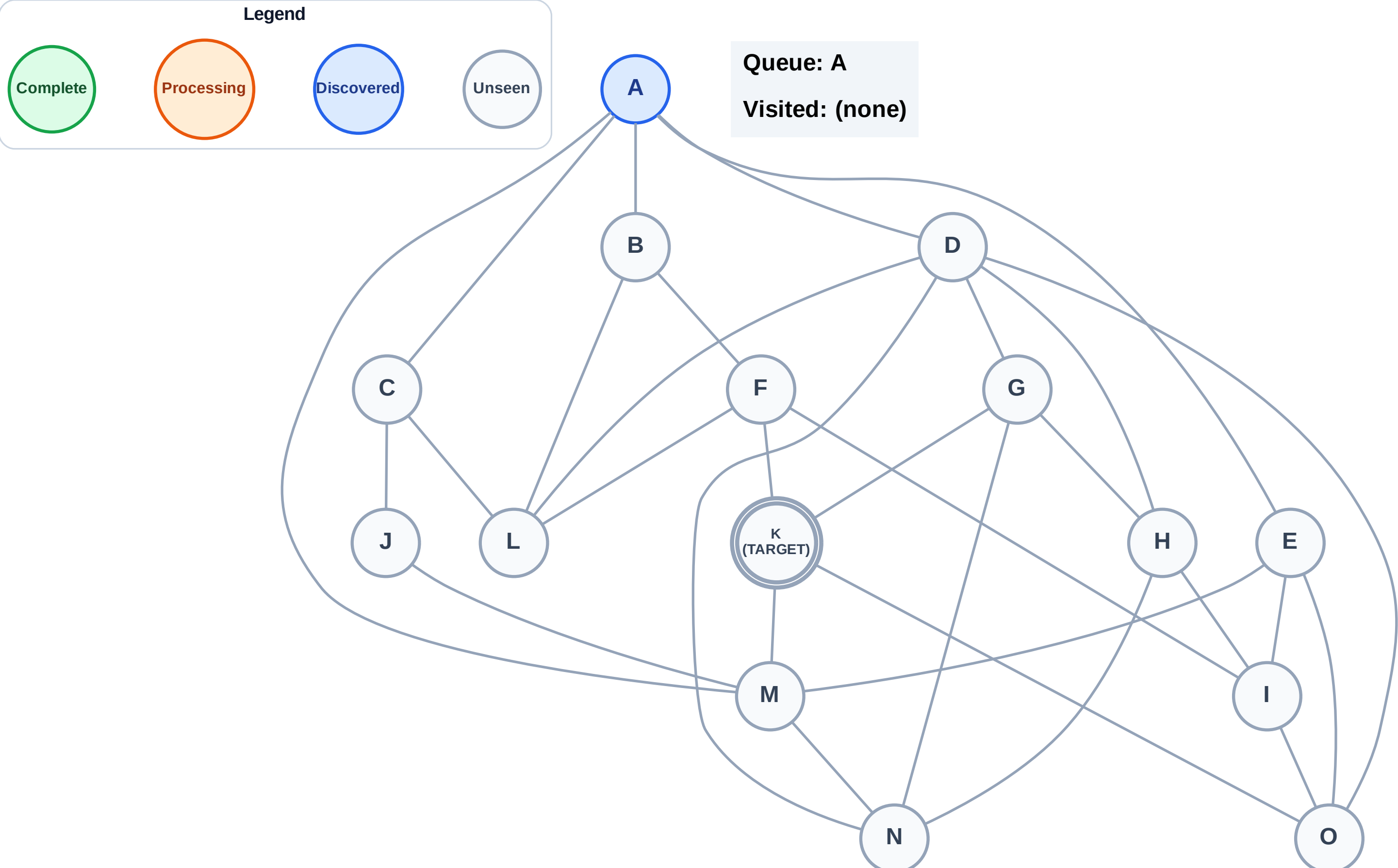


BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))



BFS Traversal

Step 2: Process node A

Legend

Complete

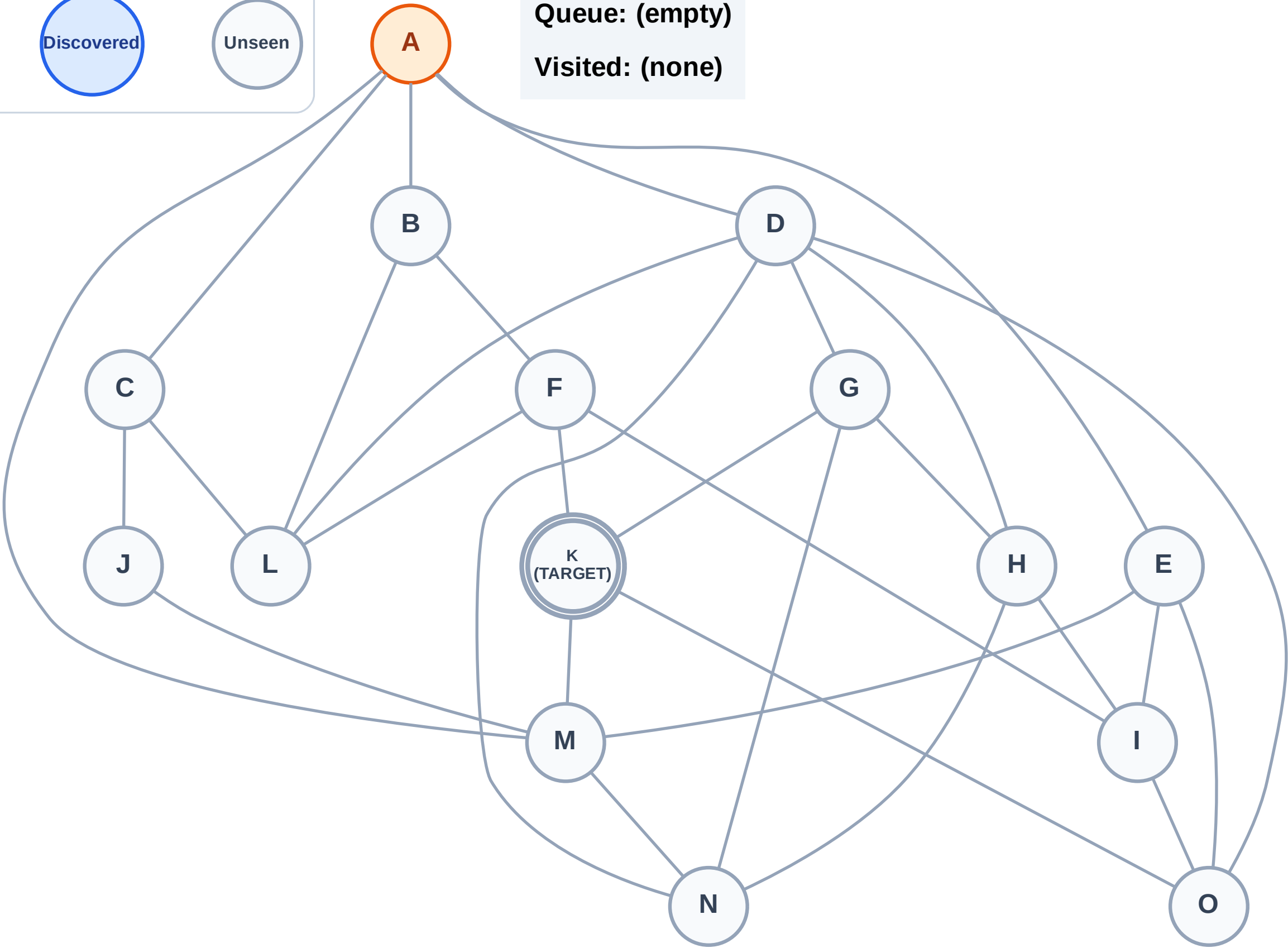
Processing

Discovered

Unseen

Queue: (empty)

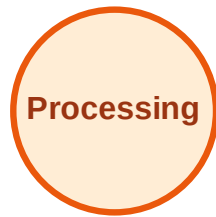
Visited: (none)



BFS Traversal

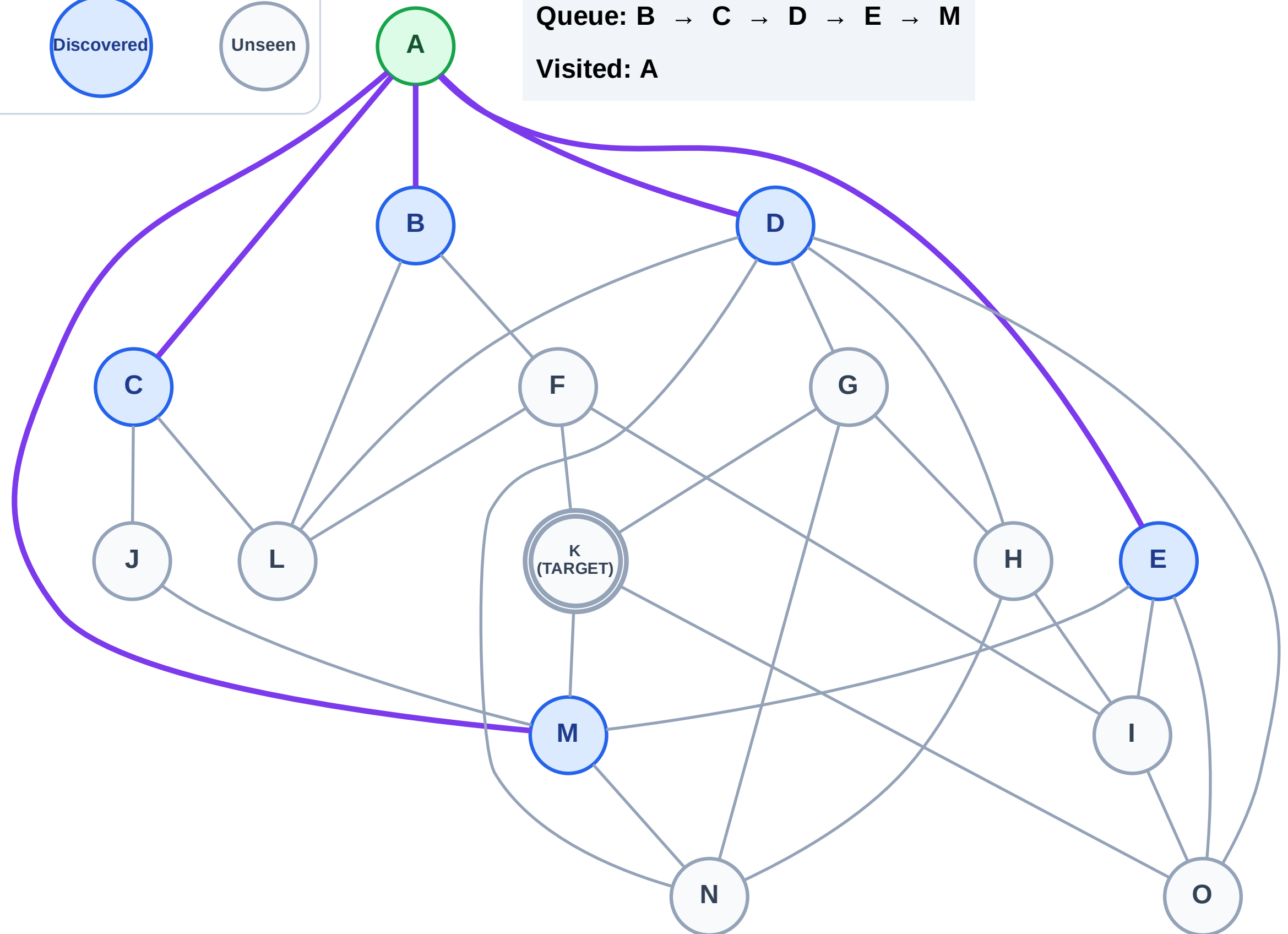
Step 3: Discover B, C, D, E, M from A

Legend



Queue: B → C → D → E → M

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

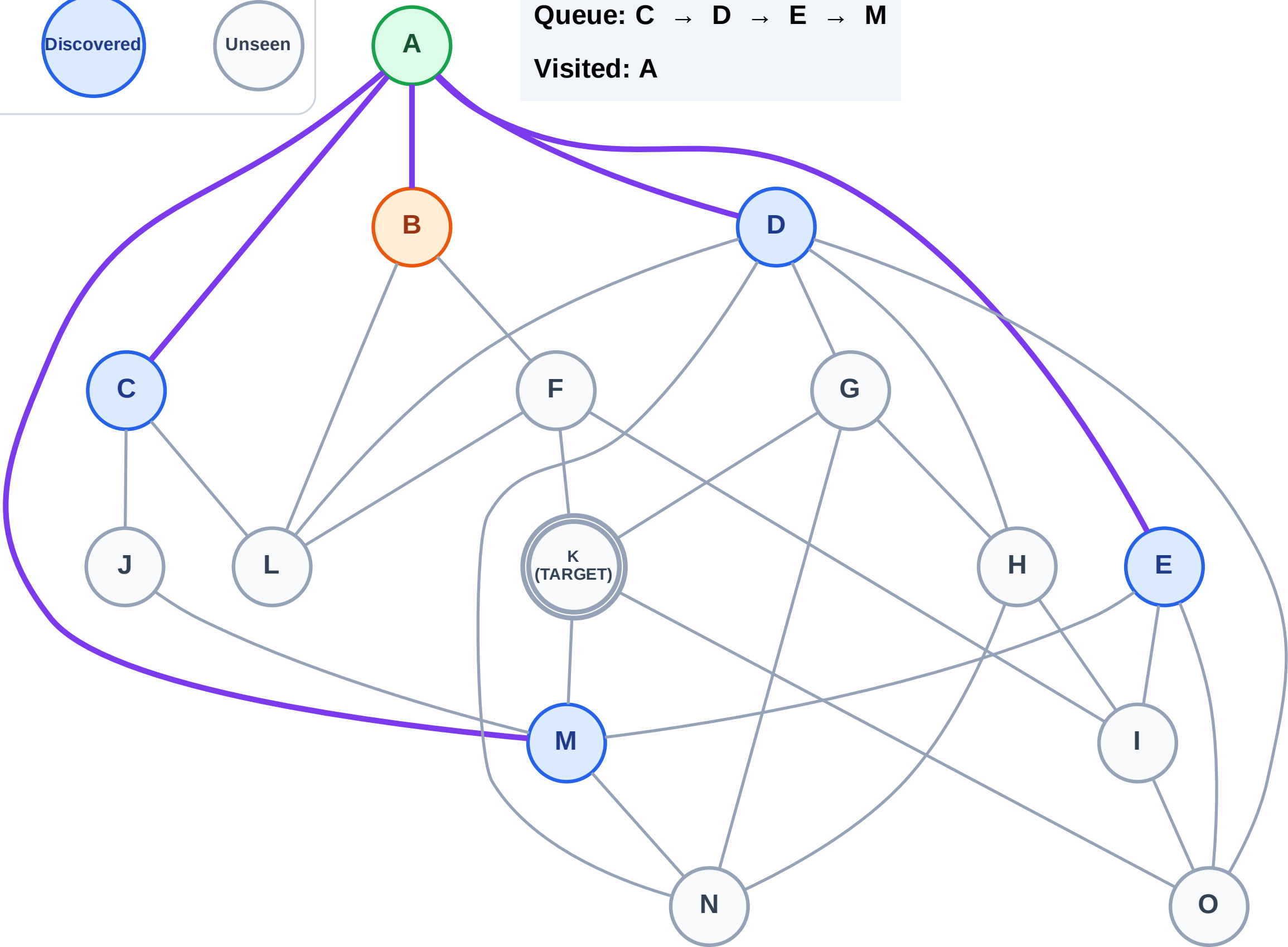
Processing

Discovered

Unseen

Queue: C → D → E → M

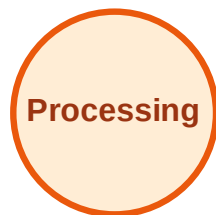
Visited: A



BFS Traversal

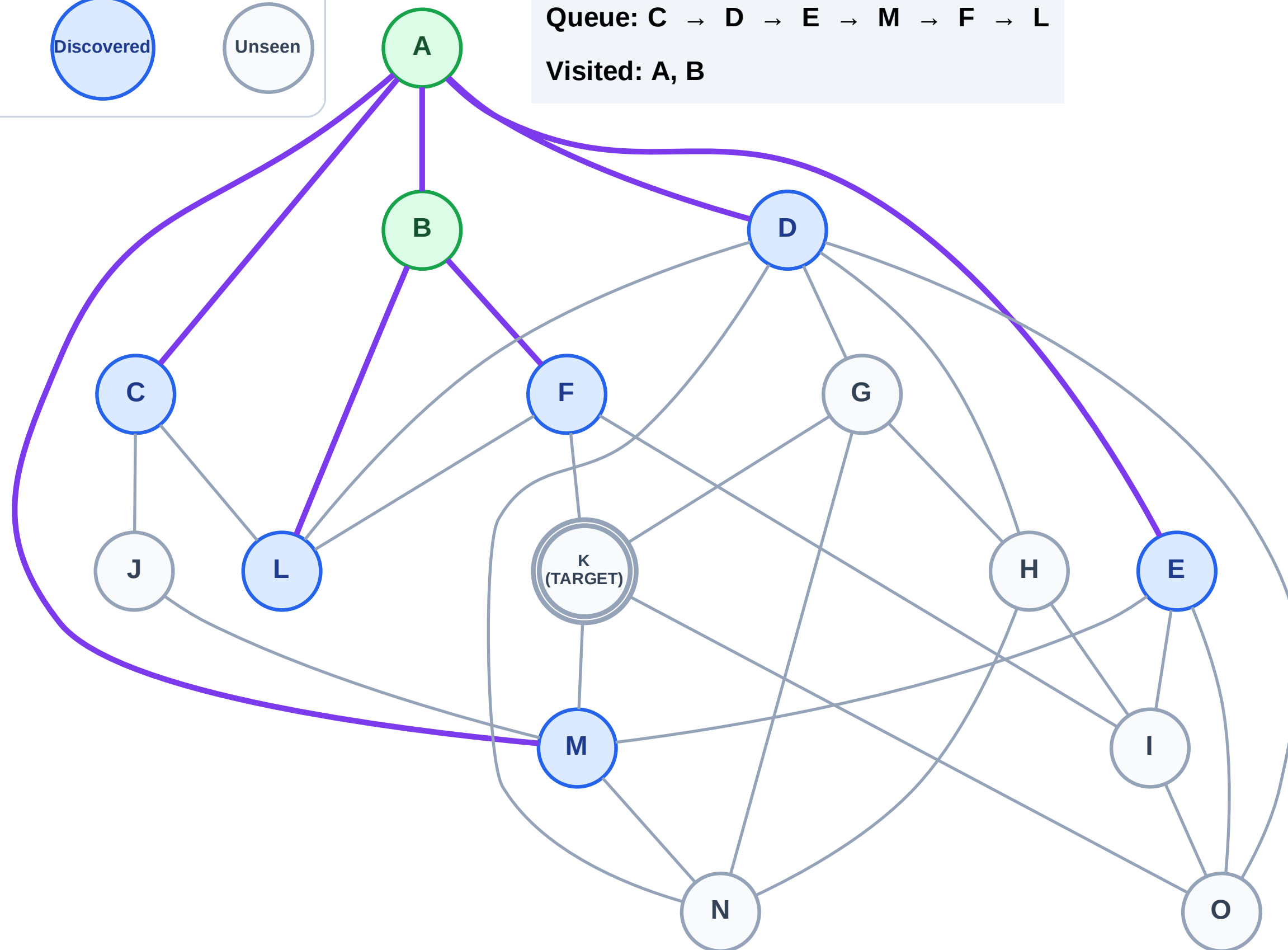
Step 5: Discover F, L from B

Legend



Queue: C → D → E → M → F → L

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

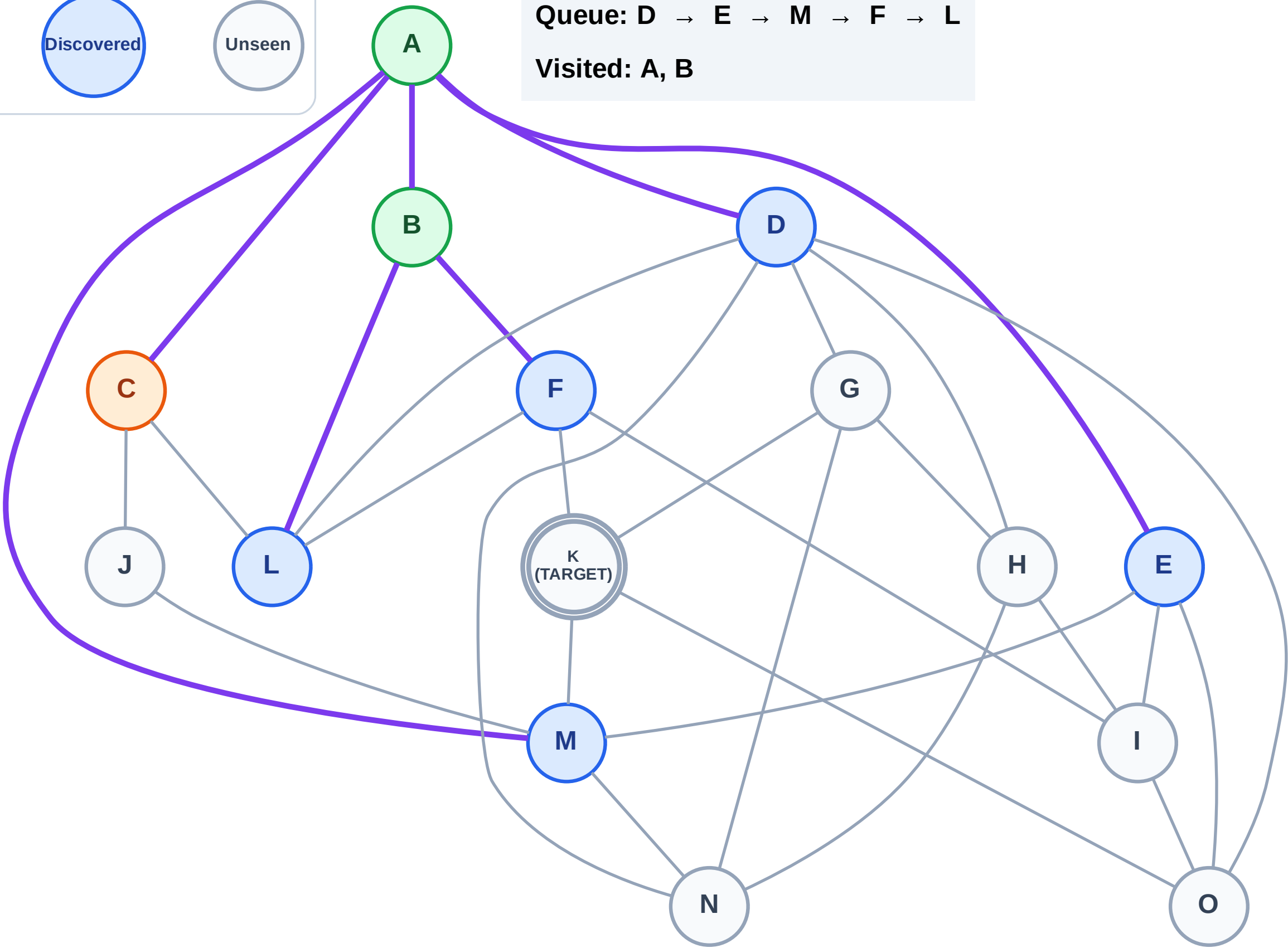
Processing

Discovered

Unseen

Queue: D → E → M → F → L

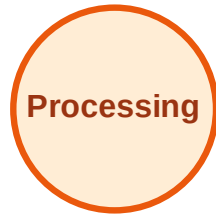
Visited: A, B



BFS Traversal

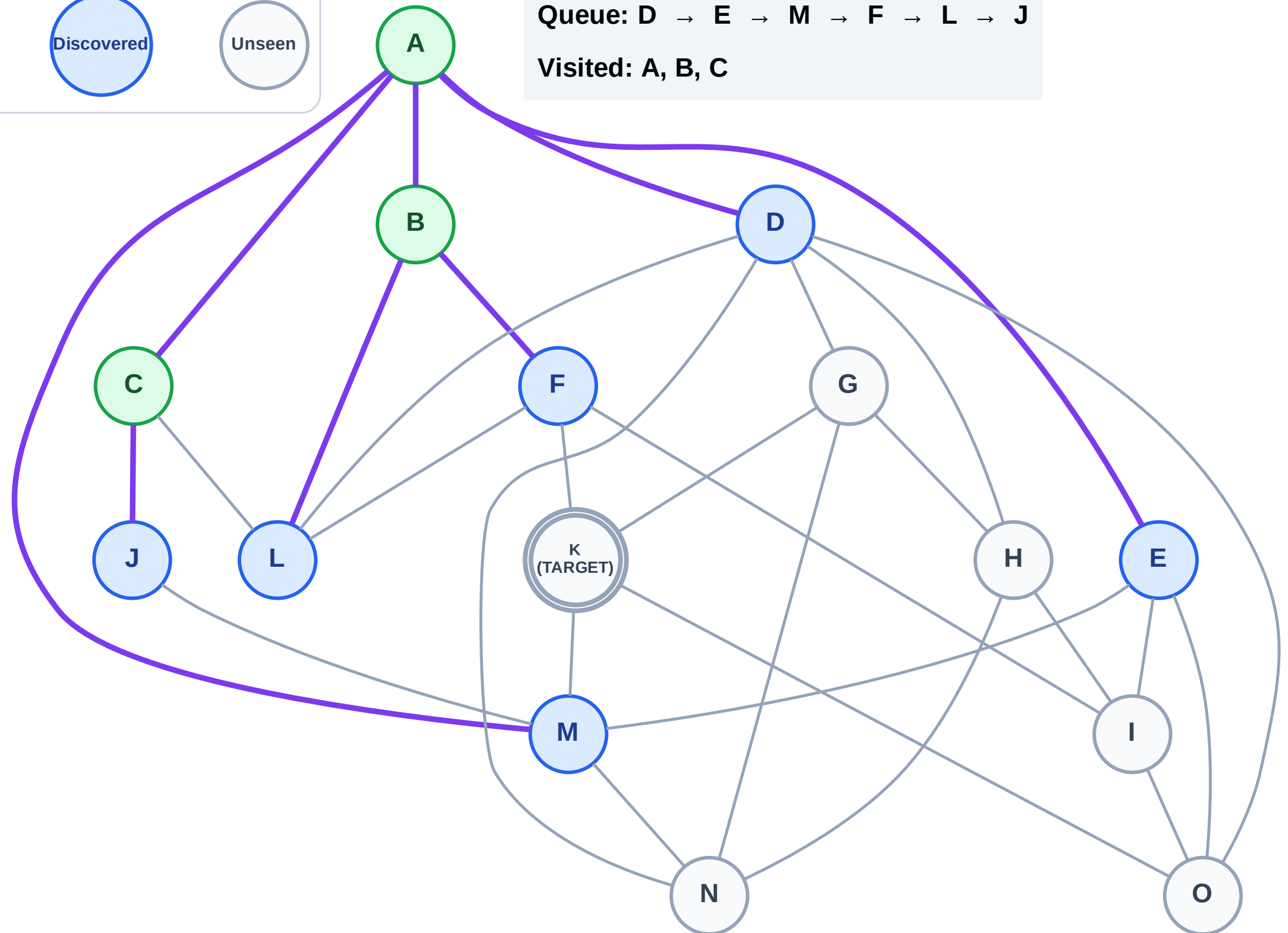
Step 7: Discover J from C

Legend



Queue: D → E → M → F → L → J

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

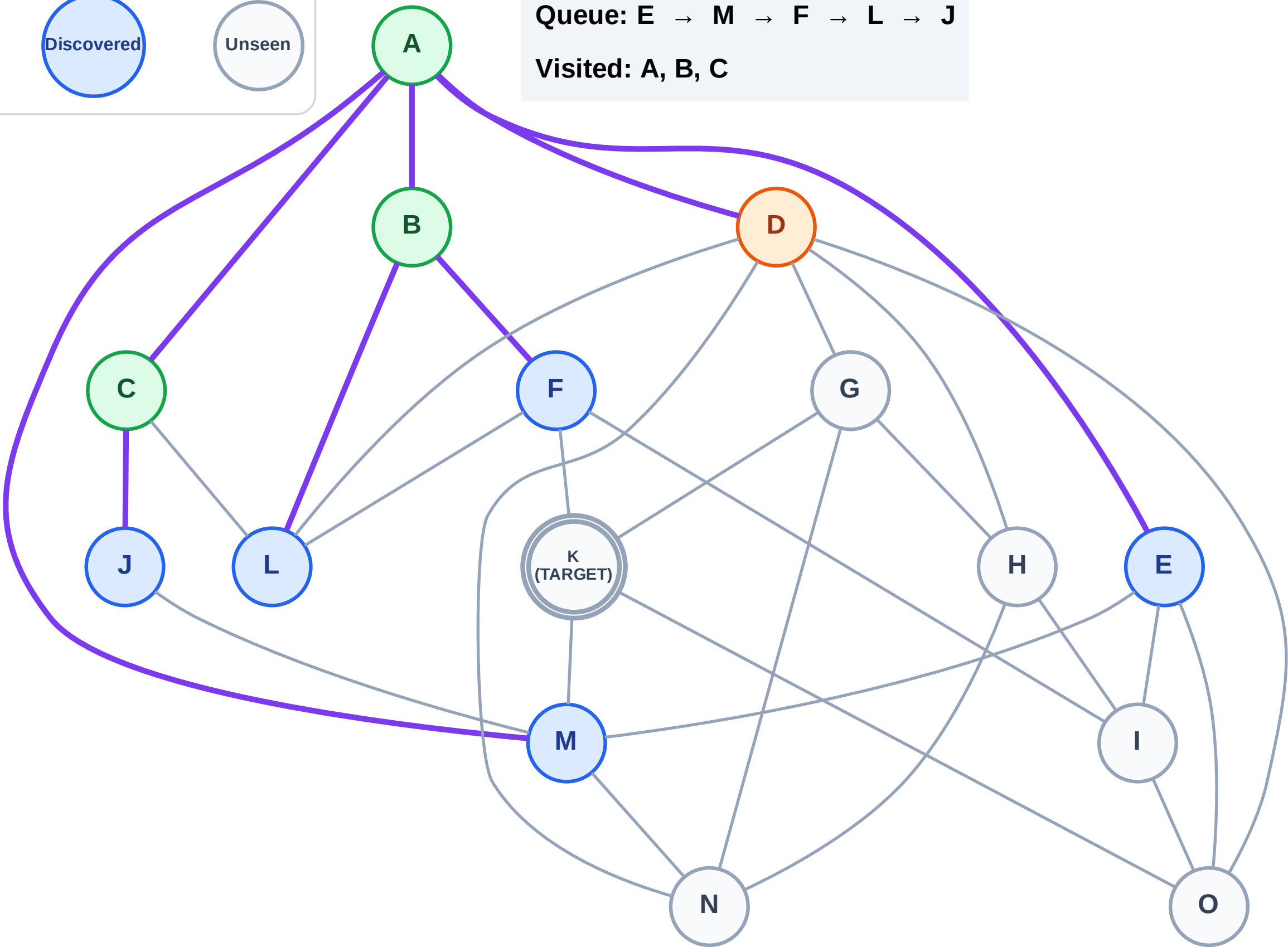
Processing

Discovered

Unseen

Queue: E → M → F → L → J

Visited: A, B, C



BFS Traversal

Step 9: Discover G, H, N, O from D

Legend

Complete

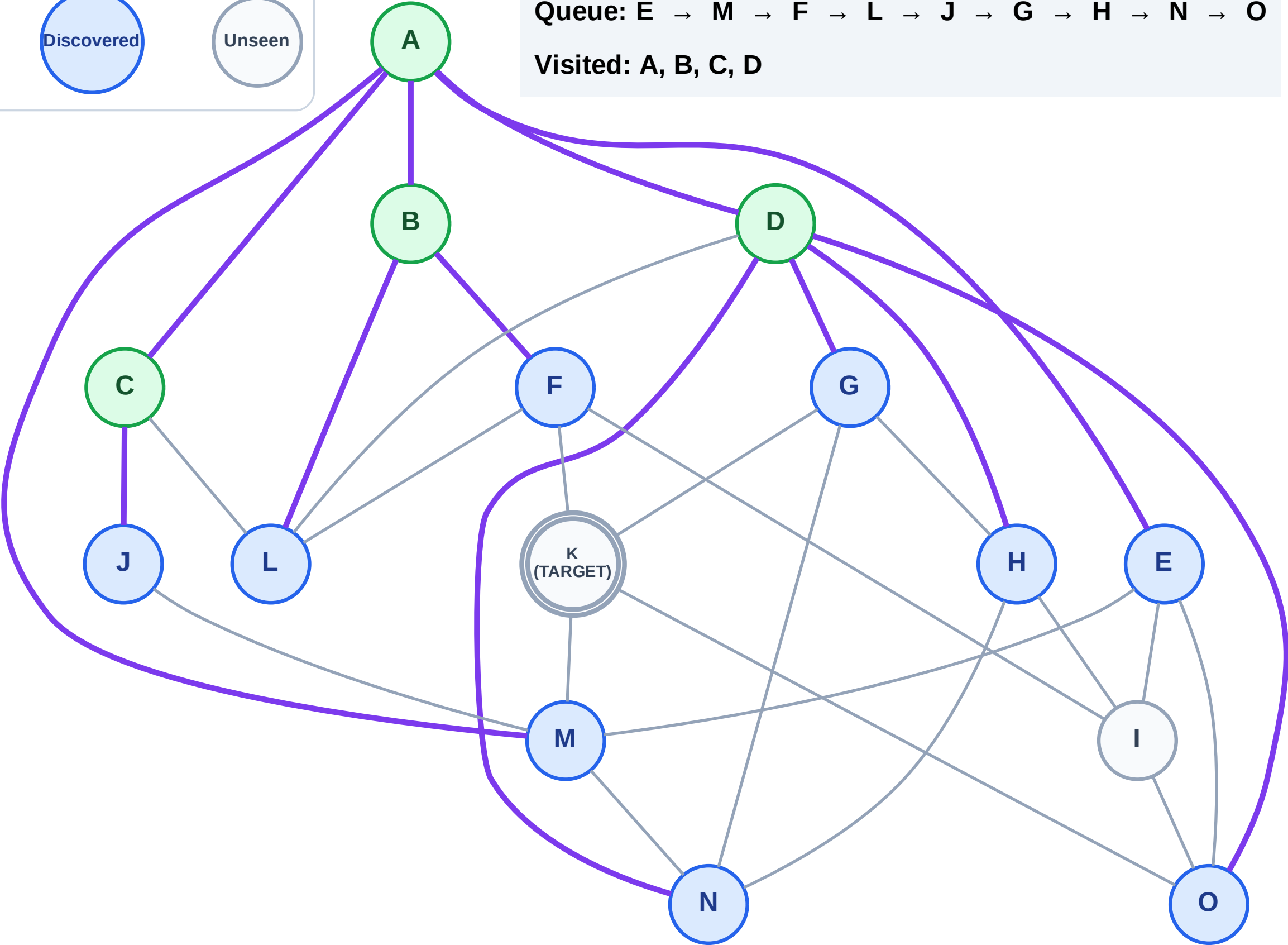
Processing

Discovered

Unseen

Queue: E → M → F → L → J → G → H → N → O

Visited: A, B, C, D



BFS Traversal

Step 10: Process node E

Legend

Complete

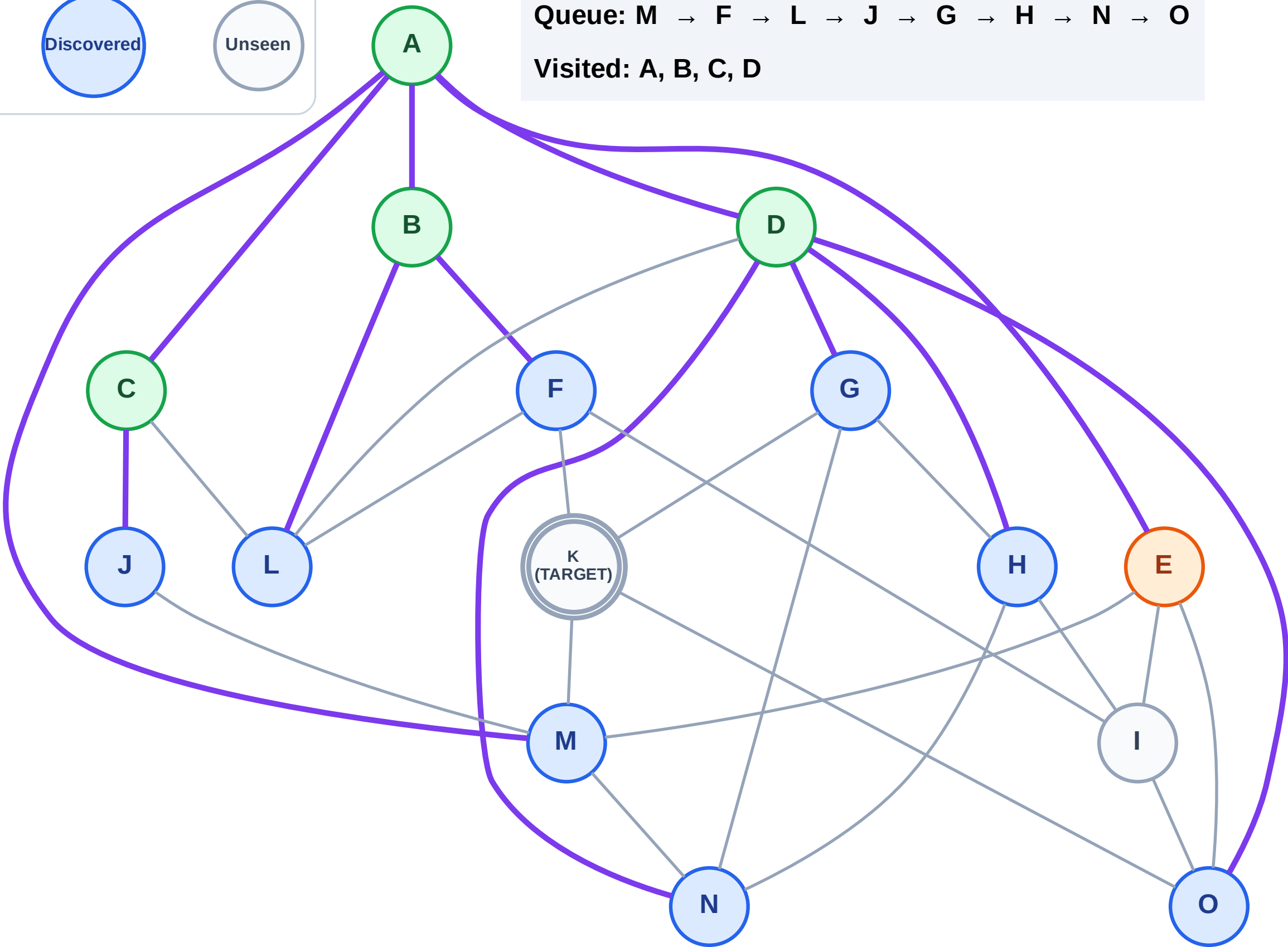
Processing

Discovered

Unseen

Queue: M → F → L → J → G → H → N → O

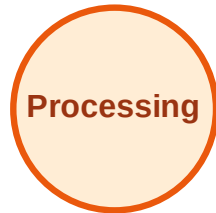
Visited: A, B, C, D



BFS Traversal

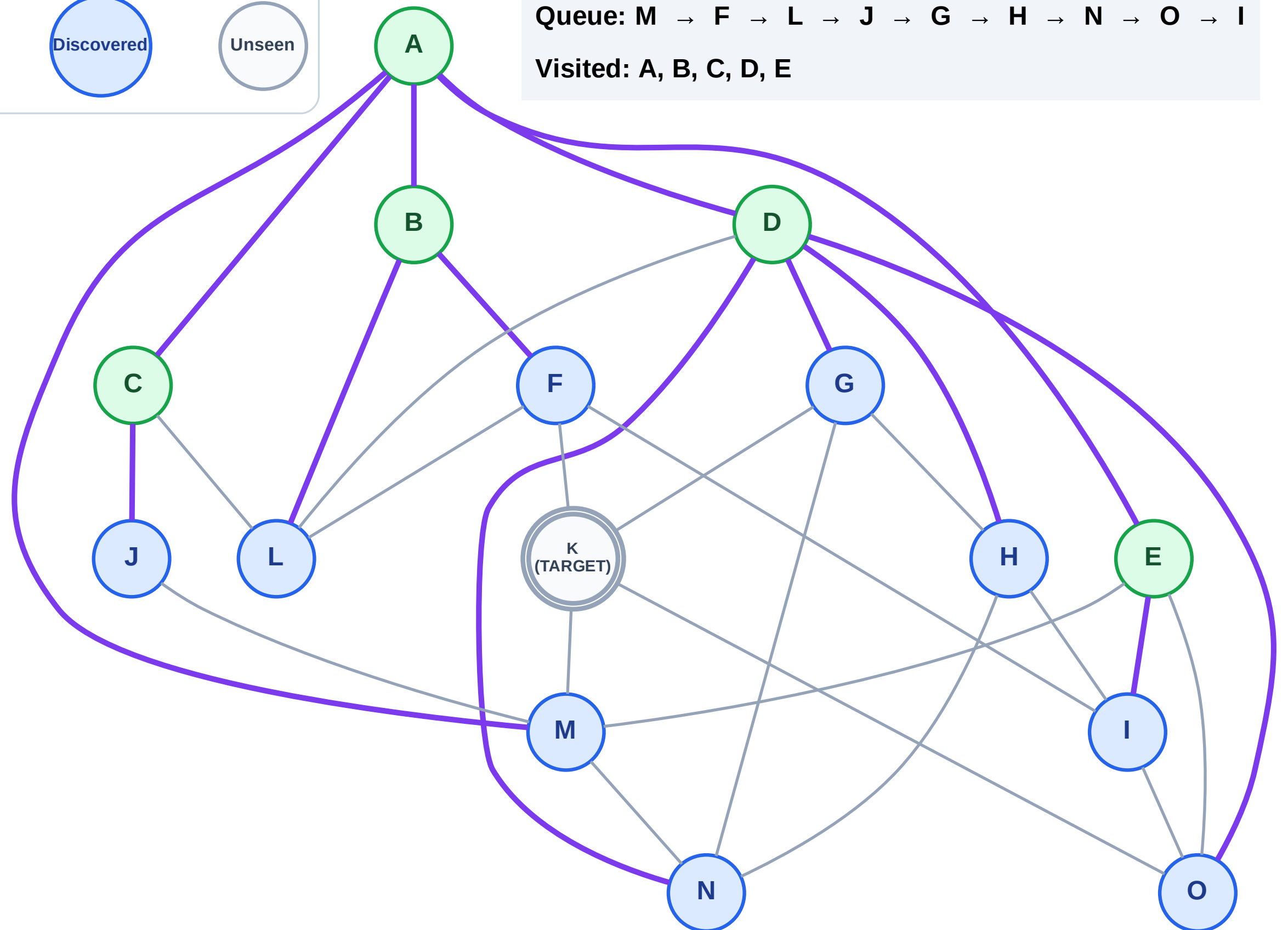
Step 11: Discover I from E

Legend



Queue: M → F → L → J → G → H → N → O → I

Visited: A, B, C, D, E



BFS Traversal

Step 12: Process node M

Legend

Complete

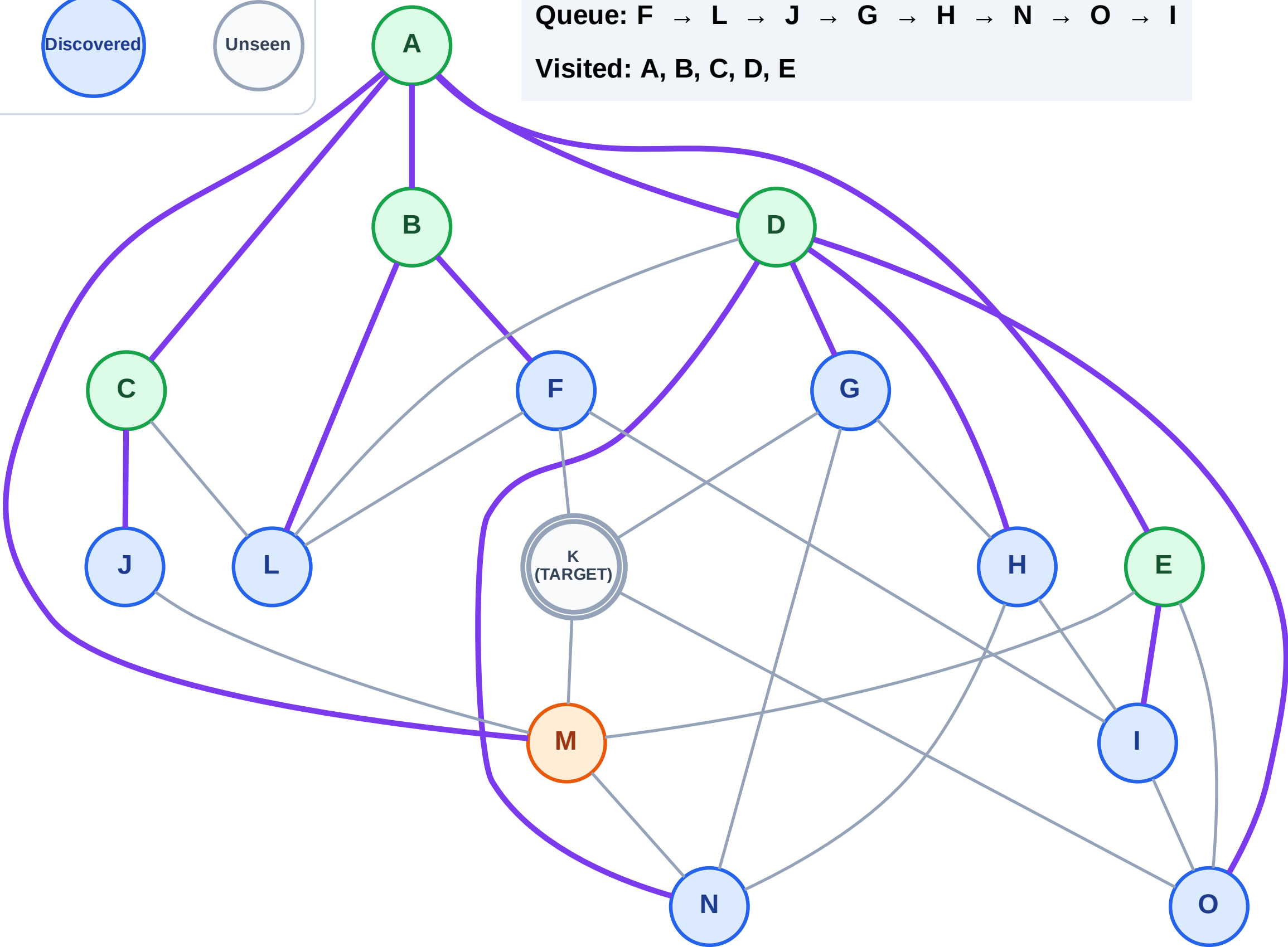
Processing

Discovered

Unseen

Queue: F → L → J → G → H → N → O → I

Visited: A, B, C, D, E



BFS Traversal

Step 13: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → M → K

Legend

Complete

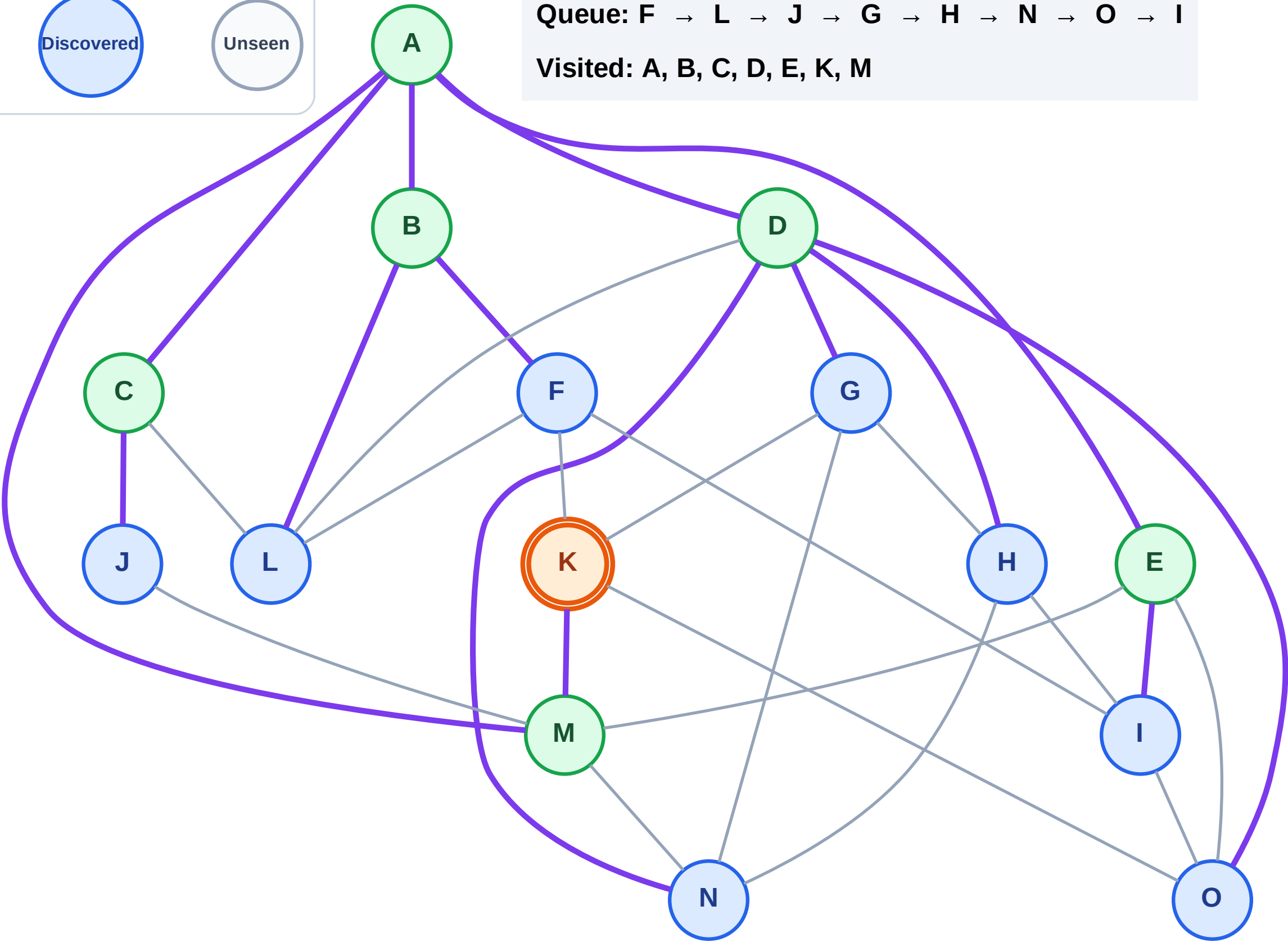
Processing

Discovered

Unseen

Queue: F → L → J → G → H → N → O → I

Visited: A, B, C, D, E, K, M



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

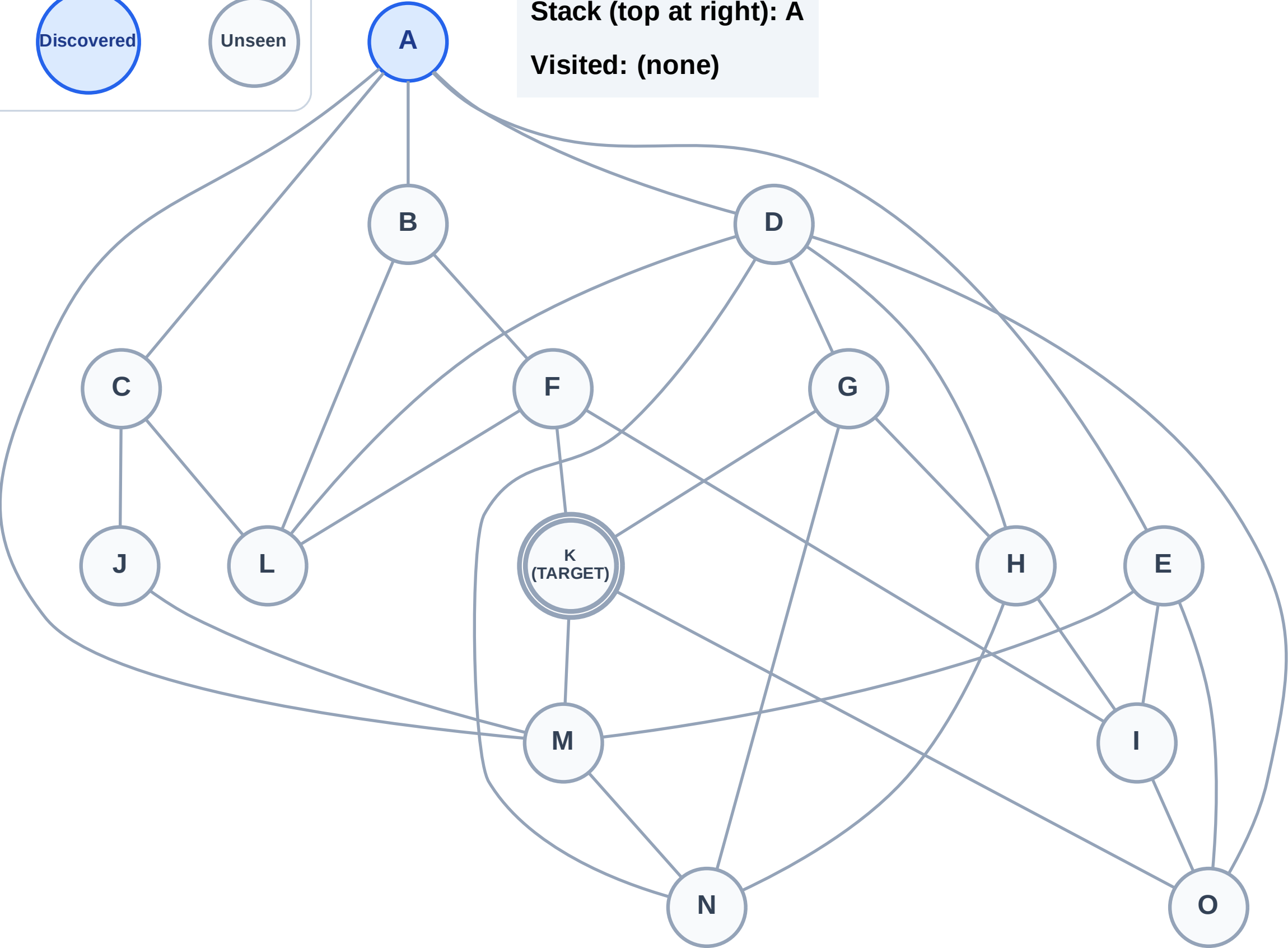
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

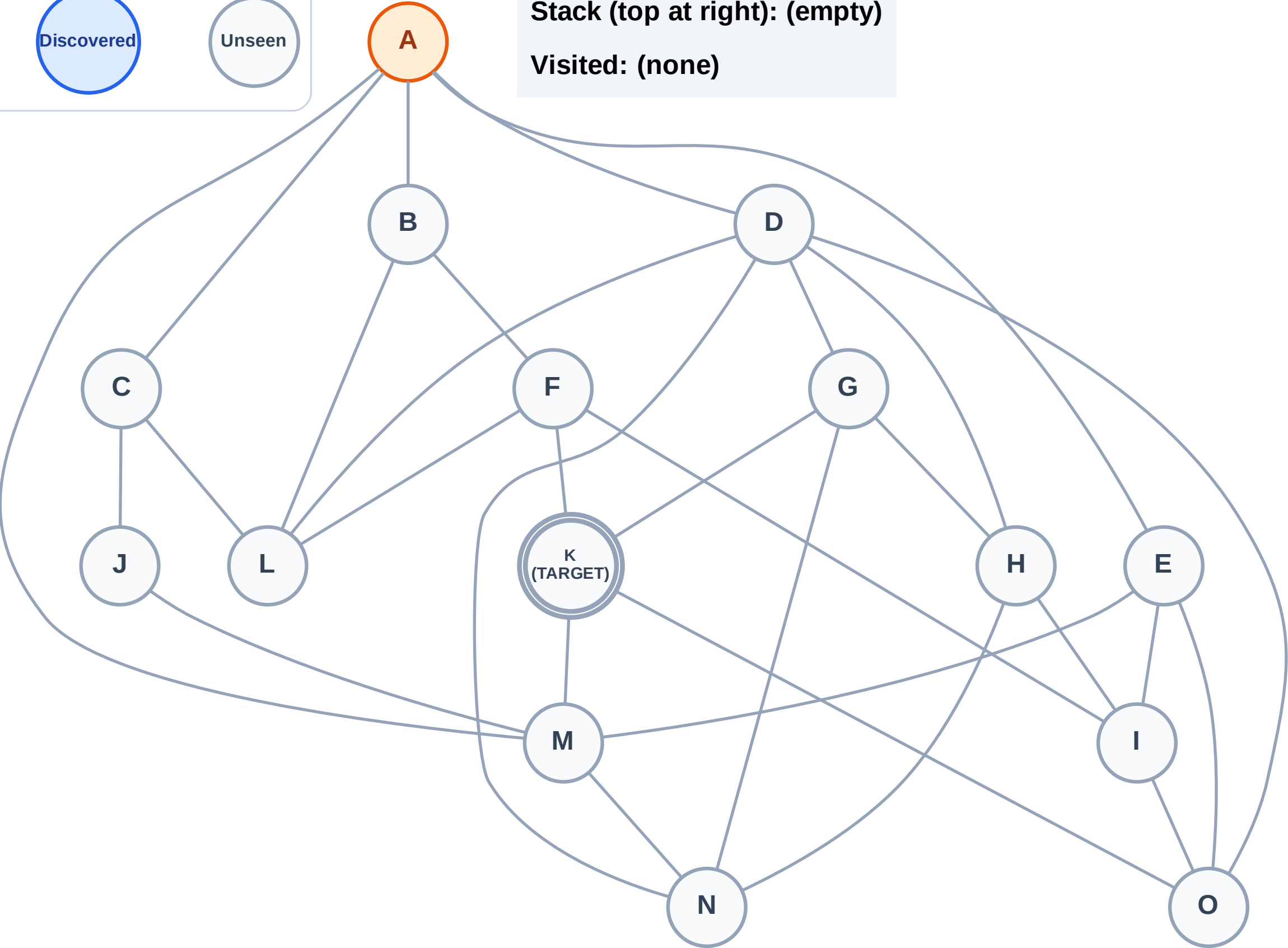
Complete

Processing

Discovered

Unseen

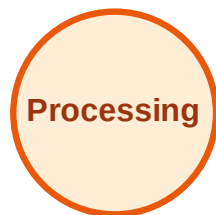
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

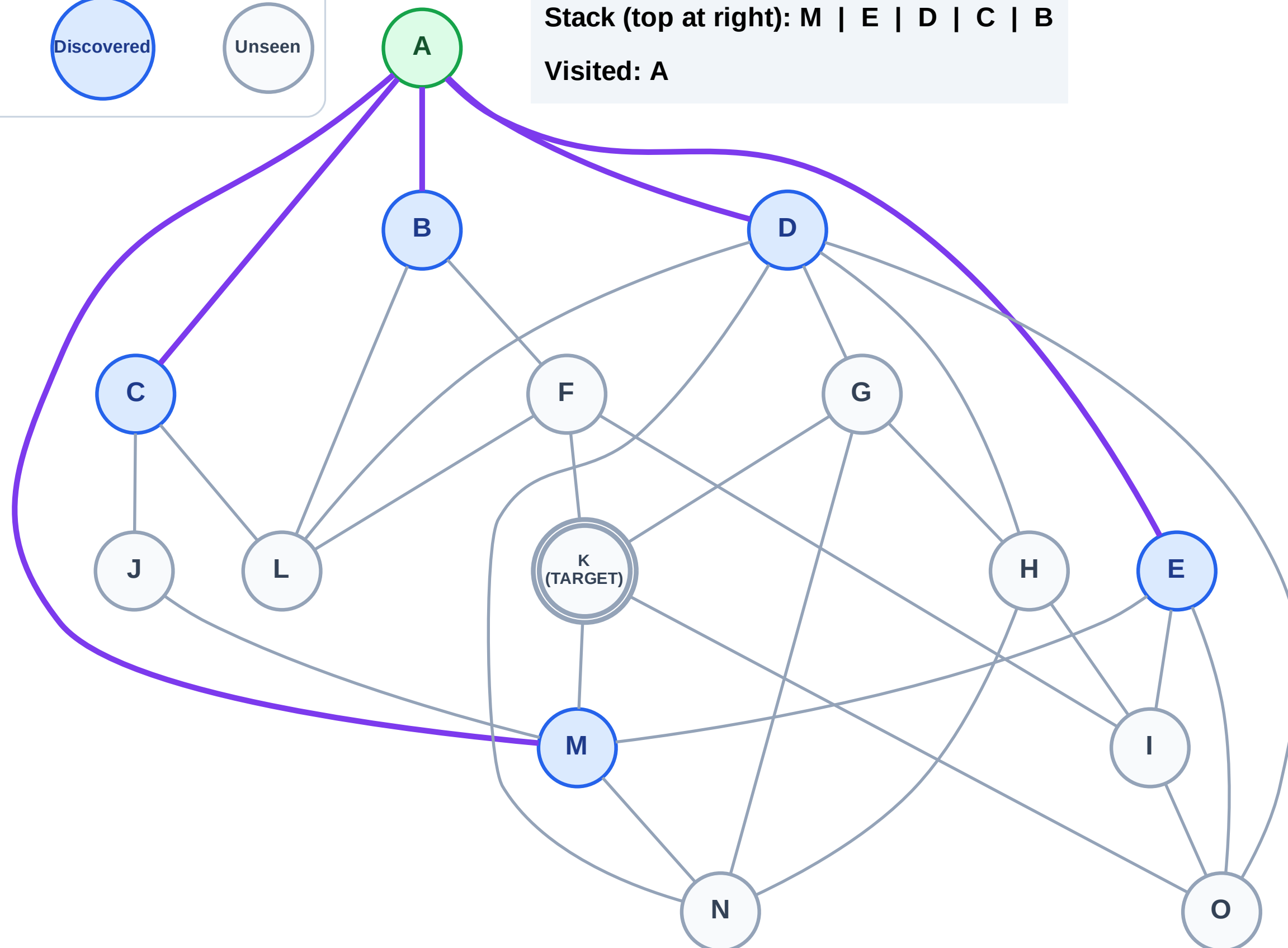
Step 3: Discover M, E, D, C, B from A

Legend



Stack (top at right): M | E | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

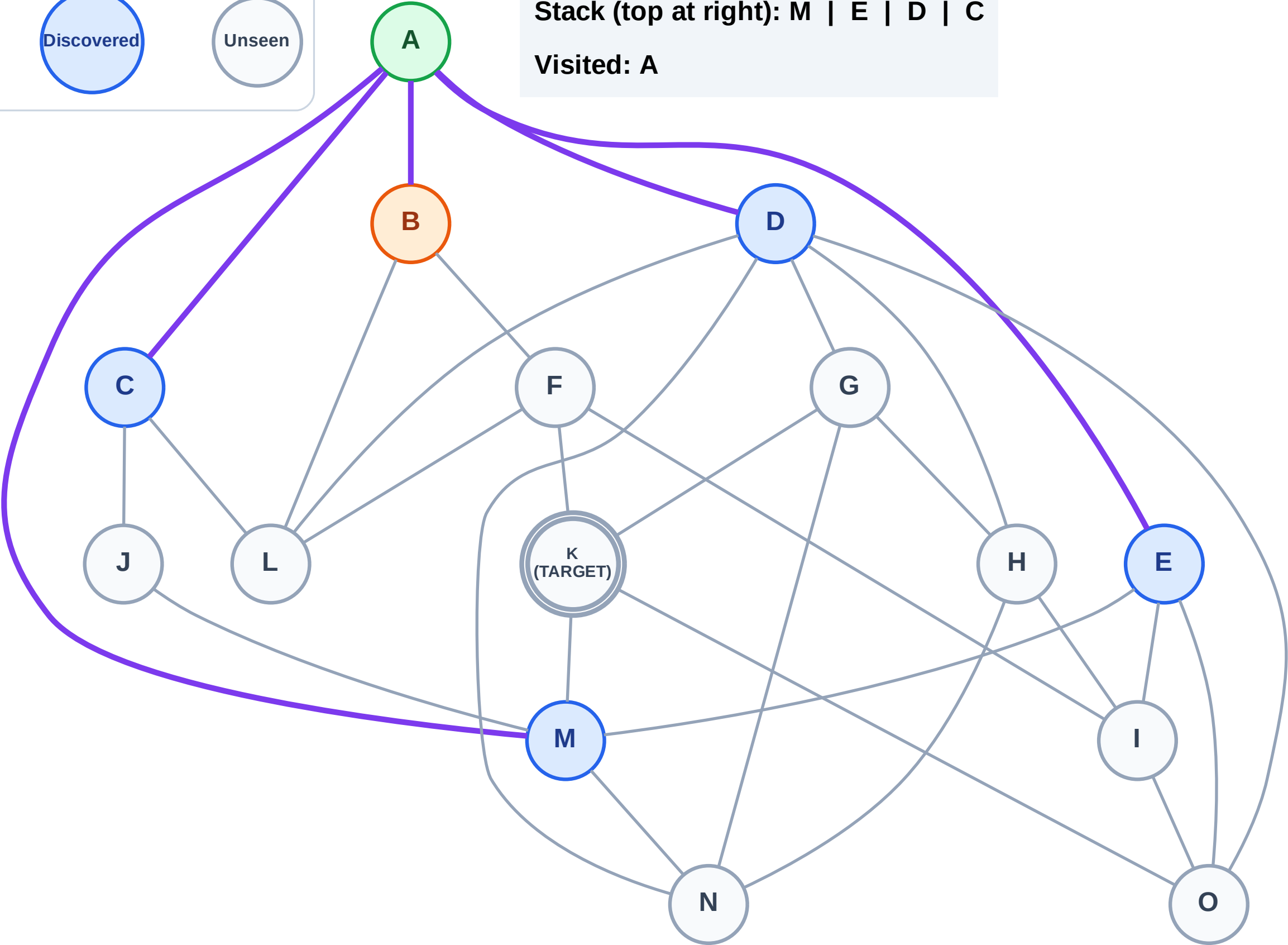
Complete

Processing

Discovered

Unseen

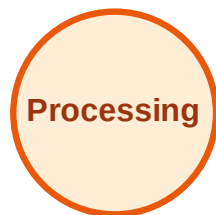
Stack (top at right): M | E | D | C
Visited: A



DFS Traversal

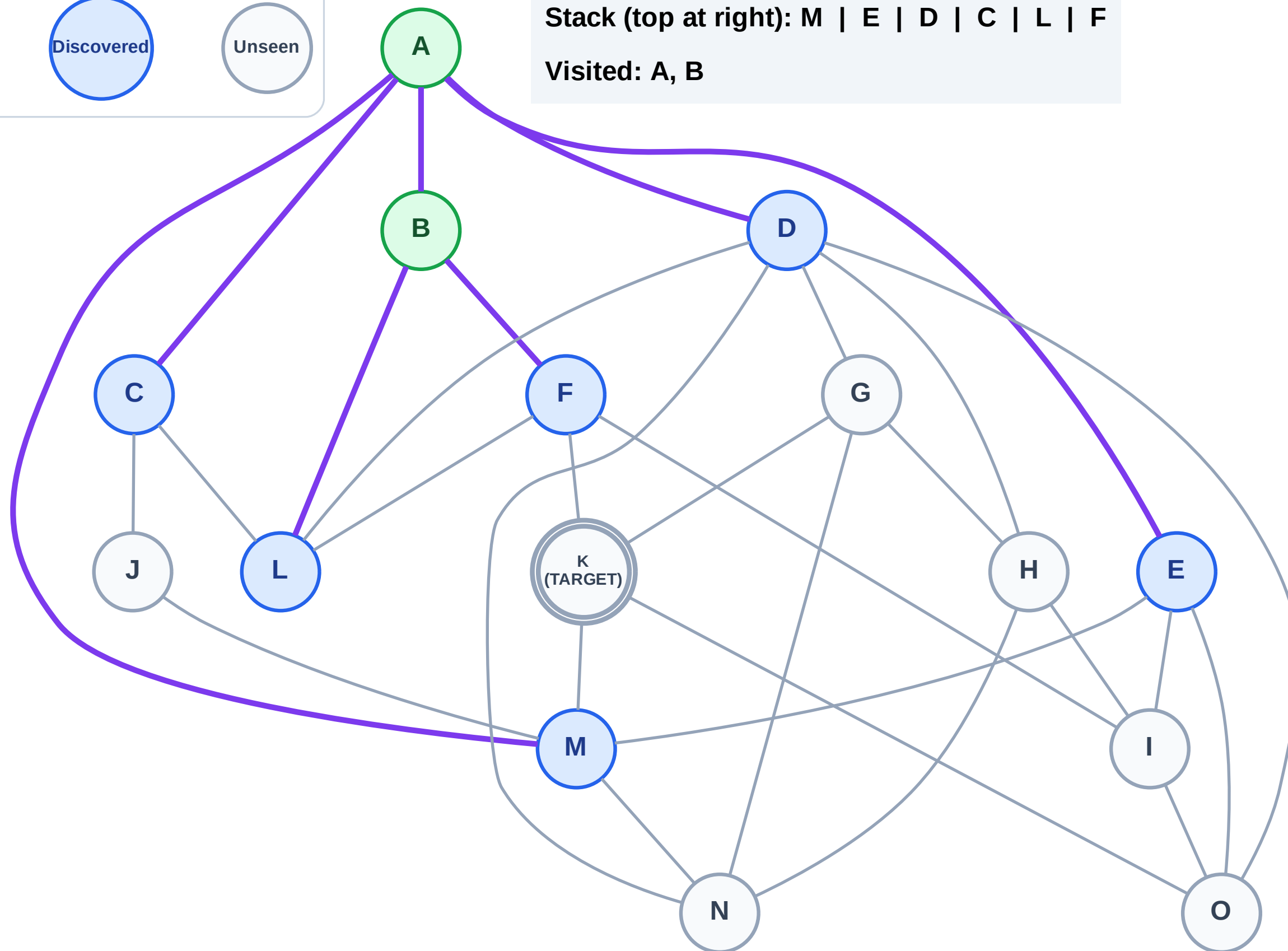
Step 5: Discover L, F from B

Legend



Stack (top at right): M | E | D | C | L | F

Visited: A, B



DFS Traversal

Step 6: Process node F

Legend

Complete

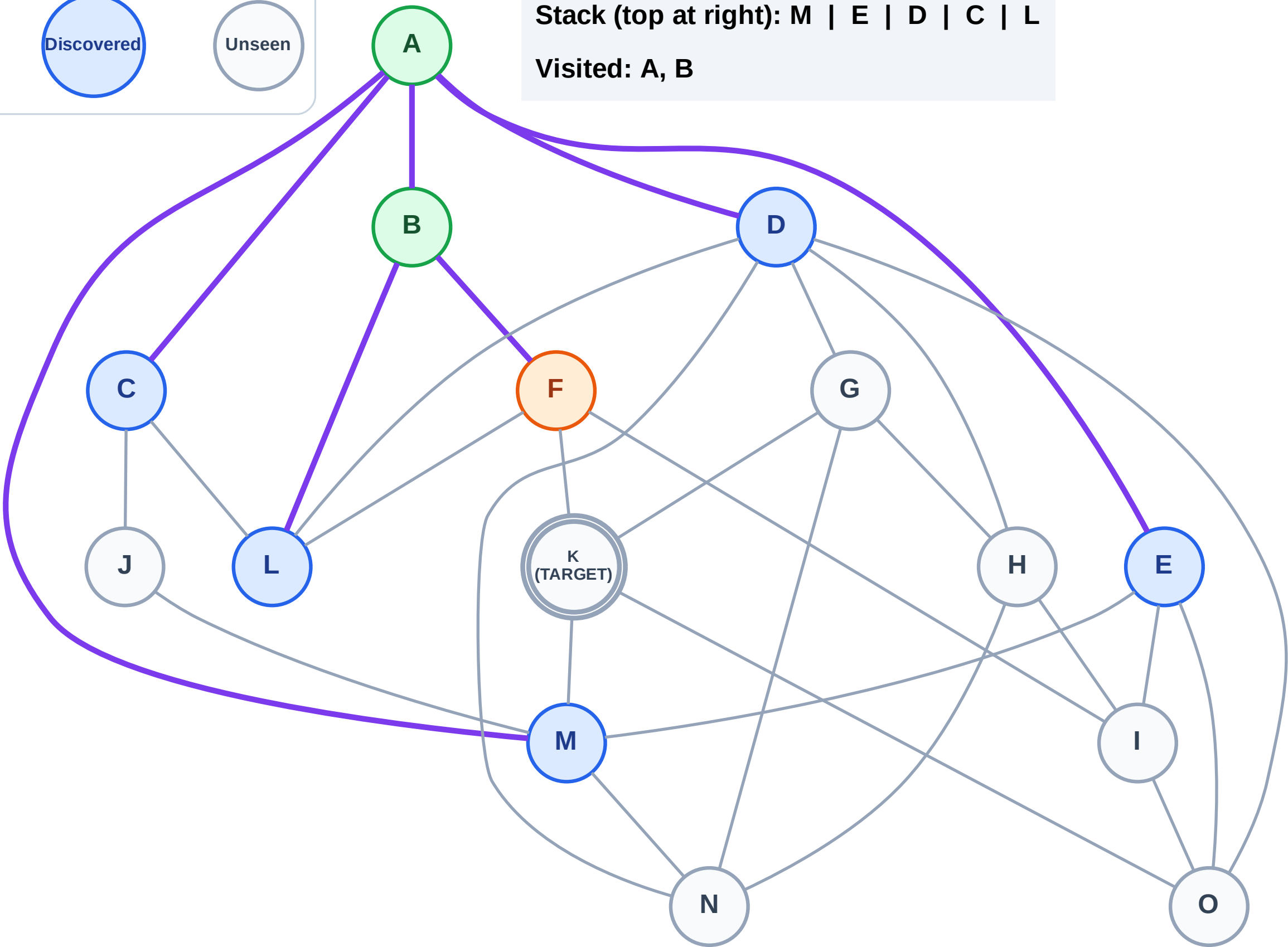
Processing

Discovered

Unseen

Stack (top at right): M | E | D | C | L

Visited: A, B



DFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → F → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): M | E | D | C | L

Visited: A, B, F, K

